

IndexError using QubitConverter to get a qubit operator

<https://github.com/qiskit-community/qiskit-nature/issues/271>

ROW 85

IndexError using QubitConverter to get a qubit operator #271

[New issue](#)[Closed](#)

#286



cposse opened on Jul 9, 2021

...

Information

- Qiskit Terra version: 0.17.4
- Python version: 3.8.10
- Operating system: Windows 10

What is the current behavior?

Script terminates with `IndexError` when converting a `ElectronicStructureProblem` to a qubit operator using `convert` method from a `QubitConverter` object.
Here is the whole traceback:

```
Traceback (most recent call last):
  File "c:\Users\poc\Documents\qiskit_scripts\test_with_error.py", line 48, in <module>
    qubit_op = converter.convert(main_op, num_particles=num_particles)
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit_nature\converters\second_quantization.py", line 100, in convert
    reduced_op = self._two_qubit_reduce(qubit_op, num_particles)
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit_nature\converters\second_quantization.py", line 110, in _two_qubit_reduce
    reduced_op = cast(PauliSumOp, two_q_reducer.convert(qubit_op))
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit\opflow\converters\two_qubit_reducer.py", line 100, in convert
    return z2_symmetries.taper(operator)
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit\opflow\primitive_ops\tapered_ops.py", line 100, in taper
    tapered_ops = self._taper(operator, self._tapering_values)
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit\opflow\primitive_ops\tapered_ops.py", line 110, in _taper
    spo = SparsePauliOp.from_list(pauli_list).simplify(atol=0.0)
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit\quantum_info\operators\sparse_pauli_op.py", line 100, in from_list
    num_qubits = len(PauliTable._from_label(obj[0][0]))
  File "C:\Users\poc\Anaconda3\envs\qiskit-0-27-psi4-jupyter\lib\site-packages\qiskit\quantum_info\operators\pauli_table.py", line 100, in _from_label
    if label[0] == '+':
IndexError: string index out of range
```

Steps to reproduce the problem

Run the example file in the README.md from <https://github.com/Qiskit/qiskit-nature>, but use a single helium atom (He) instead of H2.

In my case I slightly adjusted the example (switch to PSI4), so here's the script I used:

```
#!/usr/bin/env python
# -*- coding: utf-8 -*-

import psi4
from qiskit.algorithms.optimizers import L_BFGS_B, COBYLA
from qiskit.utils import QuantumInstance
from qiskit_nature.converters.second_quantization import QubitConverter
from qiskit_nature.drivers import Molecule, PSI4Driver
from qiskit_nature.drivers.fermionic_driver import HFMethodType
from qiskit_nature.mappers.second_quantization import ParityMapper
from qiskit_nature.problems.second_quantization.electronic import ElectronicStructureProblem

# Use PSI4, a classical computational chemistry software
# package, to compute the one-body and two-body integrals in
# electronic-orbital basis, necessary to form the Fermionic operator
# molecule = Molecule( # H2 --> works perfectly fine (E=-1.137306035696Hartree)
#     geometry=[["H", [0.0, 0.0, 0.0]], ["H", [0.0, 0.0, 0.735]]], charge=0, multiplicity=1,
# )
molecule = Molecule(geometry=[["He", [0.0, 0.0, 0.0]]], charge=0, multiplicity=1,) # He --> error
if molecule.multiplicity == 1:
    hf_method = HFMethodType.RHF
else:
    hf_method = HFMethodType.UHF
driver = PSI4Driver(
    molecule=molecule,
    basis="sto-3g", # basis sets (from fast to accurate): "sto-3g", "3-21g", "6-31g", ...
    hf_method=hf_method,
)
```

Assignees

No one assigned

Labels

[status: confirmed](#) [type: bug](#)

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Fix two qubit reduction for operators that are 2 qubits or less.](#)
qiskit-community/qiskit-nature

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Participants



```

problem = ElectronicStructureProblem(driver)

# generate the second-quantized operators
second_q_ops = problem.second_q_ops()
main_op = second_q_ops[0]

num_particles = (
    problem.molecule_data_transformed.num_alpha,
    problem.molecule_data_transformed.num_beta,
)

num_spin_orbitals = 2 * problem.molecule_data.num_molecular_orbitals

# setup the mapper and qubit converter
mapper = ParityMapper()
converter = QubitConverter(mapper=mapper, two_qubit_reduction=True)

# map to qubit operators
qubit_op = converter.convert(main_op, num_particles=num_particles)

# setup the initial state for the ansatz
from qiskit_nature.circuit.library import HartreeFock

# Ansatzes(?)/initial states (from fast to accurate): HartreeFock , ...(?), UCC, ...(?)
init_state = HartreeFock(num_spin_orbitals, num_particles, converter)

# setup the ansatz for VQE
from qiskit.circuit.library import TwoLocal

ansatz = TwoLocal(num_spin_orbitals, ["ry", "rz"], "cz")

# add the initial state
ansatz.compose(init_state, front=True, inplace=True)

# setup the classical optimizer for VQE
optimizer = L_BFGS_B(iprint=0)
# optimizer = COBYLA(dispatch=True)

from qiskit import Aer

sim_backend = Aer.get_backend("statevector_simulator")
backend = sim_backend

quantum_instance = QuantumInstance(backend=backend, optimization_level=0,)

# setup and run VQE
from qiskit.algorithms import VQE

algorithm = VQE(ansatz, optimizer=optimizer, quantum_instance=quantum_instance)

result = algorithm.compute_minimum_eigenvalue(qubit_op)
print(result.eigenvalue.real)

electronic_structure_result = problem.interpret(result)
print(electronic_structure_result)

```

The commented molecule H2 works perfectly fine, but some other molecules (He, H, H-, Ne) throw the mentioned error.

What is the expected behavior?

Convert the `ElectronicStructureProblem` to a qubit operator without error.

In case of the example script, finally print the ground state energy.

Suggested solutions

Actually I have no idea, where exactly the problem comes from.

It might even be a bug in earlier parts that propagate till `pauli_table.py`, and thus might belong to the `qiskit-nature` package.

Though I don't have enough insight to figure that out.

Any help and suggestions would be appreciated!



woodsp-ibm on Jul 9, 2021

Member ...

I moved it to Nature - you may be right that the fix may come down to logic that is in Terra, but I think Nature is the 'right' place to have this for now as things are investigated

👍 1

🔍 woodsp-ibm added type: bug on Jul 9, 2021

cposse on Jul 9, 2021 · edited by cposse

Edits ▾ Contributor Author ...

Okay.
Then the whole version info might be useful instead of only the Terra version:

Qiskit Software	Version
Qiskit	0.27.0
Terra	0.17.4
Aer	0.8.2
Ignis	0.6.0
Aqua	0.9.2
IBM Q Provider	0.14.0
Python	3.8.10 packaged by conda-forge (default, May 11 2021, 06:25:23) [MSC v.1916 64 bit (AMD64)]
OS	Windows
Qiskit Nature	0.1.3

👍

mrossinek on Jul 22, 2021

Member ...

The fact that this is an `IndexError` when accessing `label[0]` seems to suggest that the length of `label` is 0 which would mean that the number of qubits is 0. This sounds correct when simulating a single `He` atom in combination with `two_qubit_reduction=True`. We can add a check to the `QubitConverter` to skip this reduction when there are 2 or fewer qubits to begin with.

However, I don't think simulating e.g. `Ne` should cause this. Could you double check that please?

👍

🔍 mrossinek added status: confirmed on Jul 22, 2021

cposse on Jul 22, 2021

Contributor Author ...

I rechecked it and you're completely right. `Ne` works fine.
In the other cases (`H`, `H-`, `He`) the error still persists.
I tested to switch to `two_qubit_reduction=False` and then also `H`, `H-`, and `He` worked fine in agreement with your explanation.
I think the check you proposed (or even only a warning message) would be helpful for all those who start with such small examples and try out the possible options without (yet) understanding all of them in detail.

👍 1

Fix two qubit reduction for operators that are 2 qubits or less. #286

<> Code

Merged

mrossinek merged 4 commits into `qiskit-community:main` from `mrossinek:fix-271` on Jul 26, 2021

Conversation 0

Commits 4

Checks 0

Files changed 5

+54 -0



mrossinek commented on Jul 23, 2021 • edited by woodsp-ibm

Member

Summary

Fixes #271

Skip two-qubit reduction when too few qubits are available.
Also inform the user by logging a warning.



Reviewers

woodsp-ibm



manoelmarques



pbark



stefan-woerner



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[IndexError using QubitConverter to get a qub...](#)

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2 participants



mrossinek added 3 commits 4 years ago

Fix `qiskit-community#271`

Verified

2f2f3c8

Move QubitConverter test to matching source location

Verified

5a9c98e

Fix linters

Verified

d6d1293

mrossinek requested review from [manoelmarques](#), [pbark](#), [stefan-woerner](#) and [woodsp-ibm](#) as code owners 4 years ago

Add `reno`

Verified

df73128

mrossinek added the `Changelog: Bugfix` label on Jul 23, 2021

mrossinek linked an issue on Jul 23, 2021 that may be closed by this pull request

[IndexError using QubitConverter to get a qubit operator #271](#)

Closed

woodsp-ibm changed the title `Fix-#271` Fix two qubit reduction for operators that are 2 qubits or less. on Jul 23, 2021



woodsp-ibm approved these changes on Jul 23, 2021

View reviewed changes

mrossinek merged commit `99cec80` into `qiskit-community:main` on Jul 26, 2021

mrossinek deleted the `fix-271` branch 4 years ago

Anthony-Gandon pushed a commit to Anthony-Gandon/qiskit-nature that referenced this pull request on May 25, 2023

Fix two qubit reduction for operators that are 2 qubits or less. ([qis...](#))

Unverified

9a72610